11-04 第一篇

Maps: From Paper to Digital

For centuries, maps have helped people find their way. Early maps were drawn by hand on paper, leather, or even clay tablets. Explorers and traders relied on maps to navigate oceans, cross deserts, and discover new lands. As technology improved, maps became more accurate and detailed, eventually moving from paper to screens. [1] The history of mapmaking reflects humanity's growing understanding of the world and its constant desire to explore, record, and share knowledge about the places we live in and travel to.

Traditional paper maps required great skill to create. Cartographers—people who make maps—had to measure distances, observe landmarks, and calculate positions using the stars. A single map could take months or even years to complete. Despite their limitations, paper maps served explorers, soldiers, and ordinary travelers for hundreds of years. [2] The craftsmanship behind traditional maps was remarkable, and many old maps are now valued not only as historical documents but also as works of art.

The invention of satellites and computers transformed mapmaking. Satellite images provide accurate views of the earth's surface from space. Geographic Information Systems (GIS) allow scientists to layer different types of data—roads, rivers, population density—onto a single digital map. Today, anyone with a smartphone can access detailed maps of almost any place on earth. [3] Digital mapping technology has made geographic information instantly available to billions of people, changing how we travel, plan cities, and respond to emergencies.

GPS—the Global Positioning System—is one of the most widely used mapping tools today. Originally developed for military purposes, GPS now guides drivers, hikers, and delivery services. It works by communicating with a network of satellites orbiting the earth, pinpointing a user's location within a few meters. [4] The accuracy and convenience of GPS have made it an indispensable part of modern life, relied upon by individuals and industries alike for navigation and location-based services.

Despite the dominance of digital maps, paper maps have not disappeared entirely. Some hikers and sailors carry paper maps as a backup in case technology fails. Schools still teach students to read traditional maps, which helps develop spatial thinking. [5] While digital maps offer speed and convenience, the ability to read a paper map remains a valuable skill, reminding us that technology should complement—not replace—our fundamental understanding of the world around us.

选项

A. The history of mapmaking reflects humanity's growing understanding of the world and its constant desire to explore, record, and share knowledge about the places we live in and travel to. B. The craftsmanship behind traditional maps was remarkable, and many old maps are now valued not only as historical documents but also as works of art. C. Digital mapping technology has made geographic information instantly available to billions of people, changing how we travel, plan cities, and respond to emergencies. D. The accuracy and convenience of GPS have made it an indispensable part of modern life, relied upon by individuals and industries alike for navigation and location-based services. E. While digital maps offer speed and convenience, the ability to read a paper map remains a valuable skill, reminding us that technology should complement—not replace—our fundamental understanding of the world around us. F. For example, Google Maps processes over one billion kilometers of driving directions every day, making it one of the most-used apps in the world.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：空前说地图有数百年历史——手绘于纸张皮革黏土、探险者商人依赖地图、技术进步使地图更精确并从纸张走向屏幕，空后需概括地图史的意义。A的"history of mapmaking reflects humanity's growing understanding of the world"与技术进步呼应，"constant desire to explore, record, and share knowledge"与探险者商人使用地图对应。
- **[2] B**：
  - **线索**：空前说传统纸质地图制作需高超技艺——测量距离、观察地标、用星辰计算位置、一张地图可能耗时数月甚至数年、服务了探险家士兵旅行者数百年，空后需总结传统地图的价值。B的"craftsmanship behind traditional maps was remarkable"与前文技艺描述呼应，"valued not only as historical documents but also as works of art"升华其价值。
- **[3] C**：
  - **线索**：空前说卫星和电脑革新了地图制作——卫星图像提供精确视角、GIS可叠加多种数据、智能手机让任何人可访问详细地图，空后需总结数字地图技术的影响。C的"Digital mapping technology has made geographic information instantly available to billions"概括前文技术进步，"changing how we travel, plan cities, and respond to emergencies"拓展应用场景。
- **[4] D**：

- **线索**：空前说GPS——原为军事开发、现服务于驾驶者登山者快递、通过卫星网络定位精确到几米，空后需总结GPS的地位。D的"accuracy and convenience of GPS"与前文定位精度呼应，"indispensable part of modern life"与广泛应用对应。

- **[5] E**：
  - **线索**：空前说数字地图虽主导但纸质地图未消失——登山者水手携带备用、学校仍教读图培养空间思维，空后需总结纸质地图的持续价值。E的"digital maps offer speed and convenience"承认数字优势，"ability to read a paper map remains a valuable skill"与学校教学呼应，"technology should complement—not replace"是全文收尾。

- **干扰项分析**：
  - **F. For example, Google Maps processes over one billion kilometers of driving directions every day, making it one of the most-used apps in the world.**：以For example开头，为具体应用案例（谷歌地图数据）。各空均为概括或过渡句；[3]或[4]易被误选因涉及数字地图，但两空分别需数字地图技术的整体影响与GPS的综合地位，不需要谷歌地图的具体使用数据。